INCH-POUND

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7 JANUARY 2009
SUPERSEDING
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19 August 1987

DETAIL SPECIFICATION

PACKING, HYDRAULIC, LEATHER, CATTLEHIDE, MINERAL-TANNED, IMPREGNATED

INACTIVE FOR NEW DESIGN AFTER 16 JUNE 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers packing composed of impregnated, mineral tanned leather.

1.2 Classification. The leather packings are furnished in the following types and weights, as specified (see 6.2):

Type I - Impregnated with wax.
Type II - Impregnated with liquid polymer of polysulfide rubber.
Medium.
Heavy.
Extra heavy.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommend for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-311 Leather, Methods of Sampling and Testing

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The packing shall be made of cattlehide leather.

3.2 Tannage. The leather packing shall be tanned with chromium salts without the use of vegetable tanning materials.

3.3 Finish. The leather packing shall be full-grain with the flesh side smooth and free from loose flesh. It may be lightly split or skived to facilitate impregnation.

3.4 Trim. The leather packing shall be in the form of belting butt bends as specified in FED-STD-311.

3.5 Selection. When provided in the form of belting butt bends, the leather packing shall have no defects such as deep cuts, damaged grain, open scratches, open grub holes, brands, hip marks, or any other defects which effect a loss in cutting value greater than 6 percent of the total area. When purchased in the form of finished packing, the leather packing shall be free from defects.

3.6 Thickness for weights. The leather packing shall be furnished in the thickness specified for each weight shown in table I.

TABLE I. Thickness for each weight of leather packing.

Weight	Thickness	
	Minimum	Maximum
	64ths inch	64ths inch
Medium	8 (.125)	10 (.156)
Heavy	11 (.172)	13 (.203)
Extra heavy	14 (.219)	17 (.266)

3.7 Cracking. The leather packing shall not crack (see 4.5).

3.8 Tensile strength. The tensile strength shall be not less than 2,500 psi (17.24 MPa) for any one specimen (see 4.3).

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3.9 Elongation. The elongation under a load of 2,500 psi (17.24 MPa) shall be not more than 30 percent when tested with the long direction parallel to the backbone. When tested in the belly-belly direction, the elongation shall not exceed the following limits (see 4.3):

Medium - 30 percent maximum.
Heavy - 35 percent maximum.
Extra heavy - 40 percent maximum.

3.10 Shrinkage. The reduction in area of the leather packing shall be not more than 5 percent (see 4.3.2).

3.11 Porosity. The impregnated leather packing shall show no leakage of water (see 4.3.3).

3.12 Chemical composition. The leather packing shall conform to the chemical composition shown in table II (see 4.3, 4.3.1, and 4.3.5).

TABLE II. Chemical composition.

	Percent by weight			
	Type I		Type II	
	Min	Max	Min	Max
Chromic oxide 1/	4.5	-----	4.5	-----
Grease (petroleum ether extract) 2/	8	20	8	15
Impregnating wax (benzol extract) 2/	23	30	-----	-----
Chlorides as sodium chloride 1/	-----	0.30	-----	0.30
Sulfates as sulfuric acid 1/	-----	0.30	-----	0.30
Total polysulfide polymer (Thiokol) 3/	-----	-----	35	-----
Chloroform extract 3/	-----	-----	-----	15

1/ Moisture, wax, and grease-free basis.
2/ Moisture-free basis.
3/ Determined on finished leather packing. Ash and all remaining chemical tests shall be determined on a representative sample prior to application of polysulfide polymer.

3.12.1 Ash (moisture, wax, and grease-free basis). Ash shall be not more than 120 percent of the chromic oxide (see 4.3).

3.12.2 Impregnating wax. The leather packing shall be impregnated with a wax having a melting point within the range of 170 to 185 °F (76.7 to 85 °C).

4. VERIFICATION

4.1 Sampling for conformance inspection.

4.1.1 Inspection lot. All leather packing of the same type and weight presented at one time shall be considered a lot for purposes of inspection.

4.1.2 Sample for examination. A random sample of leather packing units shall be selected in accordance with table III from each inspection lot of material offered for examination of visual and dimensional characteristics. Acceptance quality level (AQL) equals 1.5 percent defective.

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TABLE III. Sampling for visual and dimensional examination.

Number of units in inspection lot	Number of units in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 and over	110	4	5

4.1.3 Sampling for tests. Samples for all physical and chemical tests of type I packing and for all physical tests and for the determination of the polysulfide rubber and chloroform extract in type II packing shall be taken from the impregnated packing. Samples for the chromic oxide, chlorides, sulfates, and ash chemical tests on type II packing shall be taken from the unimpregnated packing representative of the lot. Sampling shall be in accordance with the sample size of sampling and acceptance procedures of FED-STD-311.

4.1.4 Test specimens. Specimens shall be cut from the sample units selected as specified in 4.1.3 in accordance with figure 1. Physical test specimens shall be cut as shown on figure 2 from piece "A" shown on figure 1. Specimens for chemical analysis shall be composited from pieces "B", "C_1","C_2", and from remainder of "A" in accordance with method 6002.1 of FED-STD-311. Samples for chemical tests on unimpregnated type II packing shall be composited from pieces "B", "C_1" and "C_2".

4.2 Conformance inspection.

4.2.1 Visual and dimensional examination. Each of the sample units selected in accordance with table III shall be visually and dimensionally examined to verify conformance to this specification. Any unit in the sample containing one or more visual or dimensional defects shall be subject to rejection and if the number of defective units in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.2.2 Conformance tests. Unless otherwise specified, conformance tests shall include the tests specified in 4.3 for each type and weight.

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4.3 Tests. Tests shall be conducted in accordance with the following methods as specified in FED-STD-311 and as hereinafter specified:

Test	Method
Moisture	6221
Grease (petroleum ether boiling below 176°F (80°C)	6311
Chromic oxide, perchloric acid	6515
Soluble chlorides (on grease and wax-free sample)	6351
Soluble sulfates (on grease and wax-free sample)	6361
Total ash	6421
Tensile strength	2021
Elongation (at a stress of 2,500psi (17.24 MPa))	2021
Cracking, vise	4011
Thickness, unit	1011.1

4.3.1 Impregnating wax (benzol extract). After completing the petroleum ether extraction the same packing and extract with benzol shall be used, using a Soxhlet extractor and keeping the benzol extraction separate from the petroleum ether extraction. The benzol shall be evaporated off and dried to approximately constant weight. This extraction shall be used for the melting point of the impregnating material.

4.3.2 Shrinkage. The length and width of the specimen shall be measured to the nearest .02 inch (.508 mm) along the center lines of the specimen. The specimen shall be immersed in water at a temperature of 68 to 86° F (20 to 30°C) for 24 hours and then heated to boiling at the rate of 17 to 21.5°F (5 to 7°C) per minute and boiled for 6 minutes. The specimen shall be removed and the length and width again measured immediately. The shrinkage shall be computed as percentage of the original area.

4.3.3 Porosity. A circular porosity specimen, 3 inch (76.2 mm) in diameter made from each sample marked "A" shall support a head of 1 ft (304.8 mm) of water for 30 hours without showing porosity. For type II packing, Stoddard solvent shall be used in place of Water.

4.3.4 Chloroform extract. Approximately 10 grams of type II packing shall be extracted with chloroform using method 6311 of FED-STD-311.

4.3.5 Total polysulfide polymer. Total polysulfide polymer shall be determined as follows. Accurately weigh 0.25 to 0.30 gram of impregnated packing and place in a Parr Bomb cup. Add approximately 0.50 gram of benzoic acid, 0.2 gram of potassium perchlorate and enough sodium peroxide to make the cup two-thirds full. Intimately mix the contents and ignite the bomb. When cool, dissolve the bomb contents in distilled water, filter, neutralize to methyl orange with hydrochloric acid then add a few drops in excess, bring solution to a boil and add 10 ml of 10 percent barium chloride with constant stirring. Let the solution stand for at least 16 hours and then filter. Wash the precipitate free of chlorides, then transfer the filter paper and precipitate to a tared crucible. Ignite to constant weight. The weight of barium sulfate is converted to percent sulfur by multiplying by 0.1373, dividing by the same weight and multiplying by 100. Repeat the determination on a representative sample of unimpregnated packing. Complete the percent of total polysulfide polymer content by subtracting the sulfur content of the unimpregnated sample from that of the impregnated and multiplying the difference by 2.74. (This factor is obtained by dividing 1/0.365 where 36.5 percent is representative of the sulfur content of the polysulfide polymer.)

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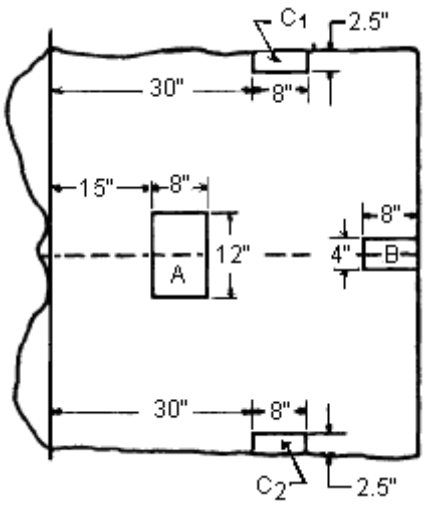

FIGURE 1. Location from which test specimens are to be taken

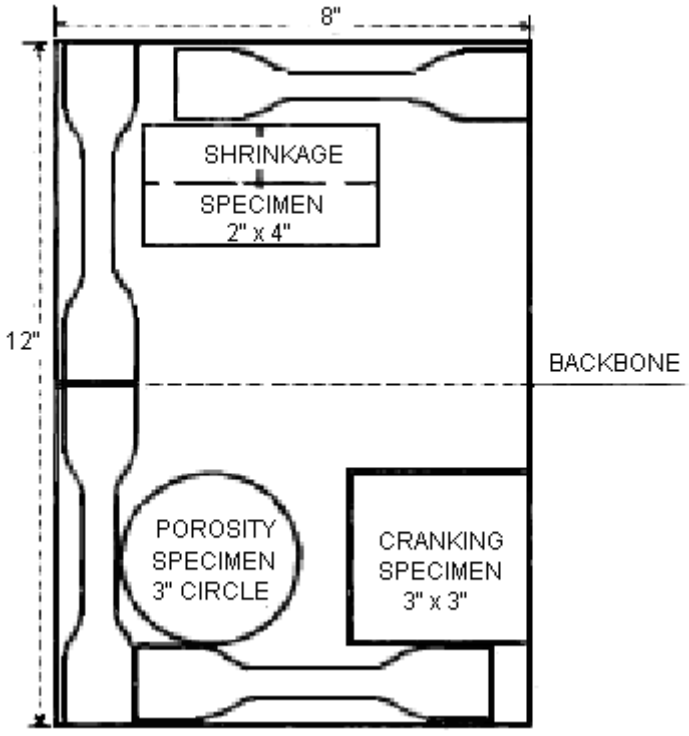

FIGURE 2. Specimen A – layout recommended

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. This material is intended primarily for hydraulic leather packing furnished in the form of belting butt bends, but may be ordered as finished packing. Type I is intended for use with hydraulic pistons operating at pressures up to 3,000 psi (20.68 MPa) at temperatures up to 155°F (68.3°C). Type II is intended for use with oil pressures up to 5,000 psi (34.47 MPa) in the temperature range from -65 to +200°F (-53.8 to +93.3°C).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and weight required (see 1.2).
(c) Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Belting butt bends
Chromium salts
Full-grain
Sulfate

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - SH

Preparing activity:
DLA – IS
(Project 5330-2009-002)

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